

Lab 5 Report: How Multilingual is Multilingual BERT?

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GitHub: https://github.com/Chu1x/TPLLMs_Lab-5-Multilinguality.git

Overview and Question Coverage

This lab fine-tunes `bert-base-multilingual-cased` for Universal Dependencies PoS tagging and evaluates cross-lingual transfer. Instead of the `Trainer` API itself, the main difficulty is the reconciliation of two tokenization schemes: UD tokens carry the gold PoS labels, while mBERT predicts over WordPiece subtokens.

The implementation is in `lab5_multilinguality.py`. The main experiment script is `scripts/run_lab5_experiments.py`.

The sections below are organized with the style of a report, roughly indicating which lab questions they answer.

Warm-up questions on the Universal Dependencies Project (Q1-Q6)

Consistent annotation is necessary because the experiment compares models across languages. If equivalent syntactic categories were annotated differently from one treebank to another, cross-lingual differences could reflect annotation mismatch rather than a real limitation of mBERT.

The provided CoNLL-U loader uses `yield` because a corpus can be read sentence by sentence without constructing the full list immediately. Calling `list(...)` then consumes the generator and stores all `(tokens, tags)` pairs for later analysis or dataset construction.

For the French Sequoia test set, I found 456 sentences and 9734 normalized tokens. The PoS distribution is very imbalanced:

Label	Count
NOUN	2125
ADP	1320
DET	1176
PUNCT	1084
VERB	781
ADJ	636
PROPN	480
ADV	417
PRON	398
AUX	345
ADP+DET	310
NUM	257
CCONJ	227
SCONJ	106
X	38
SYM	34

The plot is saved as:

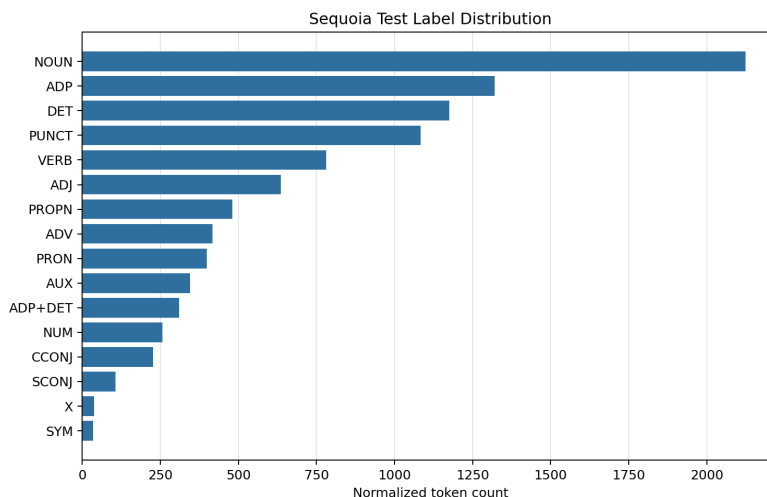


Figure 1: Sequoia label distribution

The distribution shows that common categories such as NOUN, ADP, DET, and PUNCT dominate the corpus, while X and SYM are rare. This means raw accuracy can hide poor performance on rare categories.

The Sequoia test set contains 310 multiword tokens. Examples include:

Surface form	Syntactic words	Labels
des	de les	ADP+DET
du	de le	ADP+DET
Aux	A les	ADP+DET
au	a le	ADP+DET

UD splits multiword tokens because syntactic annotation is defined over syntactic words, not necessarily over orthographic surface tokens. This makes PoS and dependency annotation more comparable across languages. For this lab, however, I keep the surface token and concatenate the child labels, for example ADP+DET. I agree with the UD decision for annotation consistency, although for mBERT input I keep surface multiword tokens to preserve the model-facing text.

UD also permits token forms containing spaces. In Sequoia, I found 13 such tokens, mostly numbers such as 500 000, 800 000, and 80 000. These spaces are removed before mBERT tokenization because the tokenizer receives a list of already-tokenized words.

The Sequoia statistics above answer the warm-up questions only. The cross-lingual experiment below uses UD_French-GSD for French, so its sentence/token counts differ from Sequoia; this is expected, not a contradiction.

mBERT Tokenization (Q7-Q9)

mBERT uses subword tokenization to reduce the number of unknown words and to share vocabulary across many languages. Rare or morphologically complex words can be represented as sequences of smaller WordPiece units instead of being mapped directly to [UNK].

For the sentence:

Pouvez-vous donner les mêmes garanties au sein de l'Union Européenne

UD syntactic-word tokenization works at the level of syntactic words. For this sentence, the UD-style token list is:

Pouvez -vous donner les mêmes garanties à le sein de l' Union Européenne

Here *Pouvez-vous* is split into the verb and clitic pronoun, *au* corresponds to *à le*, and *l'Union* is split at the elided determiner. However, in my normalised model input, UD multiword tokens are kept as surface tokens, so that *au* remains *au* and receives the merged label ADP+DET. With `bert-base-multilingual-cased`, mBERT then tokenizes the surface input as:

Po ##uve ##z - vous donner les mêmes gara ##nties au sein de l [UNK] Union Euro ##pée ##nne

The spelling is *Européenne* throughout. The difference matters because gold PoS labels are attached to UD tokens, while the model predicts one label per mBERT input token. Without an alignment rule, the label sequence and model output sequence would have different lengths.

Token/Label Alignment and Dataset Construction (Q10-Q16)

The code implements the following alignment rules:

1. UD multiword tokens are kept as surface tokens, and their child labels are concatenated, for example ADP+DET.
2. If mBERT splits one UD token into several WordPiece units, only the first subtoken receives the original label.
3. Special tokens, padding tokens, and continuation subtokens receive the label ID -100, so PyTorch loss and evaluation ignore them.

The relevant functions are:

- `normalize_ud_sentence`, which keeps multiword surface tokens and concatenates their child UPOS labels.
- `tokenize_and_align_labels`, which aligns normalized UD labels to mBERT subtokens.
- `build_label_maps`, which encodes string labels as integer IDs.
- `build_dataset_from_conllu`, which creates HuggingFace datasets from CoNLL-U files.
- `compute_accuracy_from_logits`, which masks ignored positions during evaluation.

The resulting HuggingFace Dataset contains the keys expected by mBERT: `input_ids`, `attention_mask`, and `labels`. The script uses `Dataset.from_list` to build one dataset for each train, dev, and test split.

All cross-lingual runs use one global `label2id` / `id2label` mapping. The label vocabulary is the label schema observed in the selected corpora and is used only to keep label IDs consistent across languages. It is not used to tune predictions or select results. I checked whether train/dev labels alone cover test labels; two rare composite labels appear only in test, so the full observed schema is kept to avoid unknown label IDs during evaluation.

Core implementation sketch:

```
def normalize_ud_sentence(sentence):
    for token in sentence:
        if token is multiword range:
            words.append(surface_form_without_spaces)
            labels.append(".".join(child_upos_labels))
            skip_child_tokens()
        elif token is not empty_node:
            words.append(form_without_spaces)
            labels.append(upos)

def tokenize_and_align_labels(words, labels):
    encoded = tokenizer(words, is_split_into_words=True,
                        return_offsets_mapping=True,
                        padding=True, truncation=True,
                        max_length=256)
    for start, end in encoded.offset_mapping:
        if (start, end) == (0, 0): label = -100          # special/pad
        elif start == 0: label = label2id[next_label]   # first subtoken
        else: label = -100                              # continuation

def compute_metrics(logits, labels):
    pred = argmax(logits)
```

```
mask = labels != -100
return accuracy(pred[mask], labels[mask])
```

With `truncation=True`, mBERT silently drops tokens beyond the maximum sequence length. The controlled experiment uses `max_length=256`, so truncation must be reported at that limit:

Language	Split	Truncated / Sentences	Rate	Longest tokenized length
fr	train	1 / 14450	0.0069%	562
fr	dev	0 / 1476	0%	147
fr	test	0 / 416	0%	122
es	train	0 / 14186	0%	204
es	dev	0 / 1400	0%	175
es	test	0 / 427	0%	157
en	train	0 / 12544	0%	190
en	dev	0 / 2001	0%	98
en	test	1 / 2077	0.0481%	334
ar	train	32 / 6075	0.5267%	654
ar	dev	0 / 909	0%	202
ar	test	12 / 680	1.7647%	467
zh	train	0 / 3997	0%	168
zh	dev	0 / 500	0%	134
zh	test	0 / 500	0%	155

At `max_length=256`, truncation is still rare for most languages, but Arabic test data has the highest truncation rate. This can slightly bias Arabic evaluation toward shorter examples and should be considered when interpreting Arabic as the hardest target.

Language Choice and Pires et al. Comparison (Q17)

I chose five UD treebanks:

Code	Treebank	Motivation
fr	UD_French-GSD	Romance language, Latin script
es	UD_Spanish-GSD	Romance language close to French
en	UD_English-EWT	Germanic language, Latin script
ar	UD_Arabic-PADT	Semitic language, Arabic script, richer morphology
zh	UD_Chinese-GSD	Sino-Tibetan language, Han script, no whitespace word segmentation

This selection allows comparison between close transfer, such as French to Spanish, and more distant transfer across family, script, and morphology.

Pires et al. (2019) evaluated mBERT on many languages. For NER, they used CoNLL data for Dutch, Spanish, English, and German, plus an internal 16-language dataset including Arabic, Bengali, Czech, German, English, Spanish, French, Hindi, Indonesian, Italian, Japanese, Korean, Portuguese, Russian, Turkish, and Chinese. For POS tagging, they used UD data for 41 languages: Arabic, Bulgarian, Catalan, Czech, Danish, German, Greek, English, Spanish, Estonian, Basque, Persian, Finnish, French, Galician, Hebrew, Hindi, Croatian, Hungarian, Indonesian, Italian, Japanese, Korean, Latvian, Marathi, Dutch, Norwegian Bokmaal and Nynorsk, Polish, European and Brazilian Portuguese, Romanian, Russian, Slovak, Slovenian, Swedish, Tamil, Telugu, Turkish, Urdu, and Chinese.

This is broad coverage, but not a controlled linguistic sample. Family, script, morphology, corpus size, domain, pretraining resource availability, and annotation quality vary simultaneously, which makes it hard to isolate the source of cross-lingual transfer. Strong transfer may therefore reflect relatedness, shared script, lexical overlap, similar word order, or larger training data rather than multilinguality alone.

Corpus Size as a Confound (Q18)

Language	Split	Sentences	Tokens
fr	train	14450	344961
fr	dev	1476	34664
fr	test	416	9738
es	train	14186	375031
es	dev	1400	36461
es	test	427	11733
en	train	12544	201963
en	dev	2001	24788
en	test	2077	24740
ar	train	6075	191871
ar	dev	909	25987
ar	test	680	24201
zh	train	3997	98614
zh	dev	500	12665
zh	test	500	12010

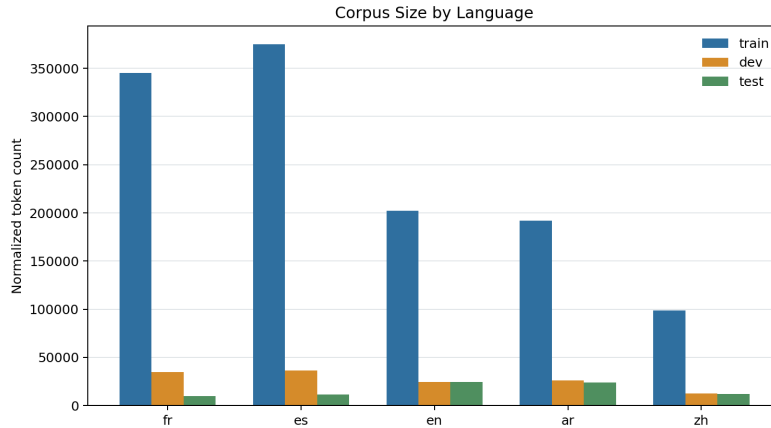


Figure 2: Corpus size by language

The training corpora differ substantially in size. French and Spanish have many more training tokens than Chinese, while Arabic has fewer sentences but long, token-dense examples. Corpus size can therefore confound conclusions about multilinguality that better results may reflect more supervised examples rather than better cross-lingual representations.

Training Setup and Metrics (Q14-Q16, Q20)

All models use `bert-base-multilingual-cased` with a token-classification head. The script reports PoS accuracy during training via `compute_accuracy_from_logits`. This is important because loss alone is not easy to interpret as tagging quality, and the metric must ignore all -100 positions.

Controlled experiment configuration:

Setting	Value
Checkpoint	<code>bert-base-multilingual-cased</code>
Initialization	Every model starts from the same mBERT checkpoint
Languages	<code>fr, es, en, ar, zh</code>
Train/dev/test caps	1000 / 300 / 300 sentences

Setting	Value
<code>max_length</code>	256
Epochs	2
Batch size	4
Learning rate	2e-5
Weight decay	0.01
Random seed	42
Metric	token accuracy, excluding -100 labels
Label map	one global 138-label vocabulary shared by all models

The `max_length = 256` setting reduces memory pressure but increases truncation relative to the default 512-token limit. Arabic was trained on CPU because pilot runs showed memory pressure on Apple MPS. All other settings are shared.

Cross-Lingual Experiment Results (Q19)

The full-corpus experiment can be launched with `make full`. Since that is slow on a laptop, I used the controlled setting:

`make controlled`

This caps all languages to 1000 train, 300 dev, and 300 test sentences, uses `max_length = 256`, and trains for 2 epochs. This is not identical to training on the full corpora, but it controls corpus size more directly and is more practical on a laptop.

The controlled matrix is saved to:

`outputs/lab5_controlled/cross_lingual_accuracy_matrix.csv`

Pilot run, using only 200 training sentences, 100 dev sentences, 100 test sentences, and 1 epoch:

Train	Test	fr	es	en	ar	zh
fr		0.265	0.192	0.154	0.343	0.280
es		0.261	0.319	0.212	0.342	0.280
en		0.206	0.254	0.180	0.375	0.347
ar		0.004	0.006	0.012	0.010	0.017
zh		0.184	0.195	0.167	0.332	0.314

This pilot matrix is only a technical check. The models are undertrained, the sample sizes are tiny, and the Arabic row is affected by the MPS memory issue. It should not be used for the final linguistic conclusions.

Controlled 5 x 5 matrix:

Train	Test	fr	es	en	ar	zh
fr		0.964	0.916	0.853	0.553	0.647
es		0.929	0.953	0.862	0.549	0.634
en		0.839	0.835	0.947	0.446	0.628
ar		0.549	0.568	0.465	0.901	0.497
zh		0.634	0.607	0.625	0.398	0.931

The mean in-language accuracy is 0.939, while the mean cross-lingual accuracy is 0.652. Best transfer directions are **es** -> **fr** at 0.929, **fr** -> **es** at 0.916, and **en** -> **fr** at 0.839. These are all between Latin-script European languages. The strongest in-language versus best-transfer gaps are for Arabic and Chinese. Arabic drops from 0.901 in-language to 0.568 best transfer (**ar** -> **es**), and Chinese drops from 0.931 to 0.634 (**zh** -> **fr**).

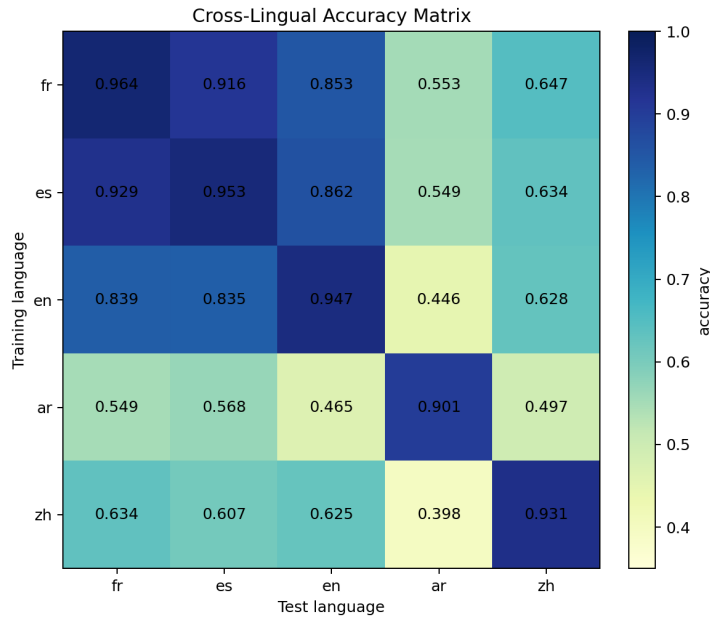


Figure 3: Controlled cross-lingual accuracy heatmap

Arabic is the hardest target. Averaging only cross-lingual scores by test language gives: **fr** 0.738, **es** 0.732, **en** 0.701, **zh** 0.602, and **ar** 0.486. French and Spanish transfer very well to each other, consistent with shared Romance family membership and Latin script. The Chinese row is asymmetric: training on Chinese transfers moderately to French, Spanish, and English, but those languages transfer less well to Chinese than to each other.

Hyperparameter Fairness and Transfer Analysis (Q20-Q21)

Using the same hyperparameters makes the comparison more controlled, but it is not automatically fair. Different languages have different corpus sizes, sentence lengths, morphology, and tokenization behavior. A fair protocol should fix the main hyperparameters in advance, use comparable training budgets, and only tune on development data.

Transfer is strongest between languages that share script, family, and morphosyntactic properties, especially French and Spanish. Transfer is weaker when script, morphology, and tokenization differ, especially when Arabic is the target language. The Chinese row is asymmetric: training on Chinese transfers moderately to French, Spanish, and English, but those languages transfer less well to Chinese than they do to each other. The matrix should therefore be interpreted against typological distance, script, segmentation, and corpus properties rather than as a single scalar measure of multilinguality.

References

- Pires, Telmo, Eva Schlinger, and Dan Garrette. 2019. *How multilingual is Multilingual BERT?* ACL. <https://aclanthology.org/P19-1493/>
- Google Research publication page: <https://research.google/pubs/how-multilingual-is-multilingual-bert/>